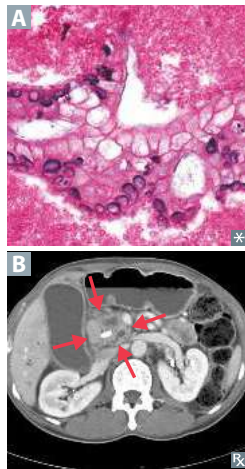


Pancreatic adenocarcinoma



Very aggressive tumor arising from pancreatic ducts (disorganized glandular structure with cellular infiltration **A**); often metastatic at presentation, with average survival ~ 1 year after diagnosis. Tumors more common in pancreatic head **B** (lead to obstructive jaundice). Associated with CA 19-9 tumor marker (also CEA, less specific). Most common genomic abnormality is *KRAS*-activating mutation.

Risk factors:

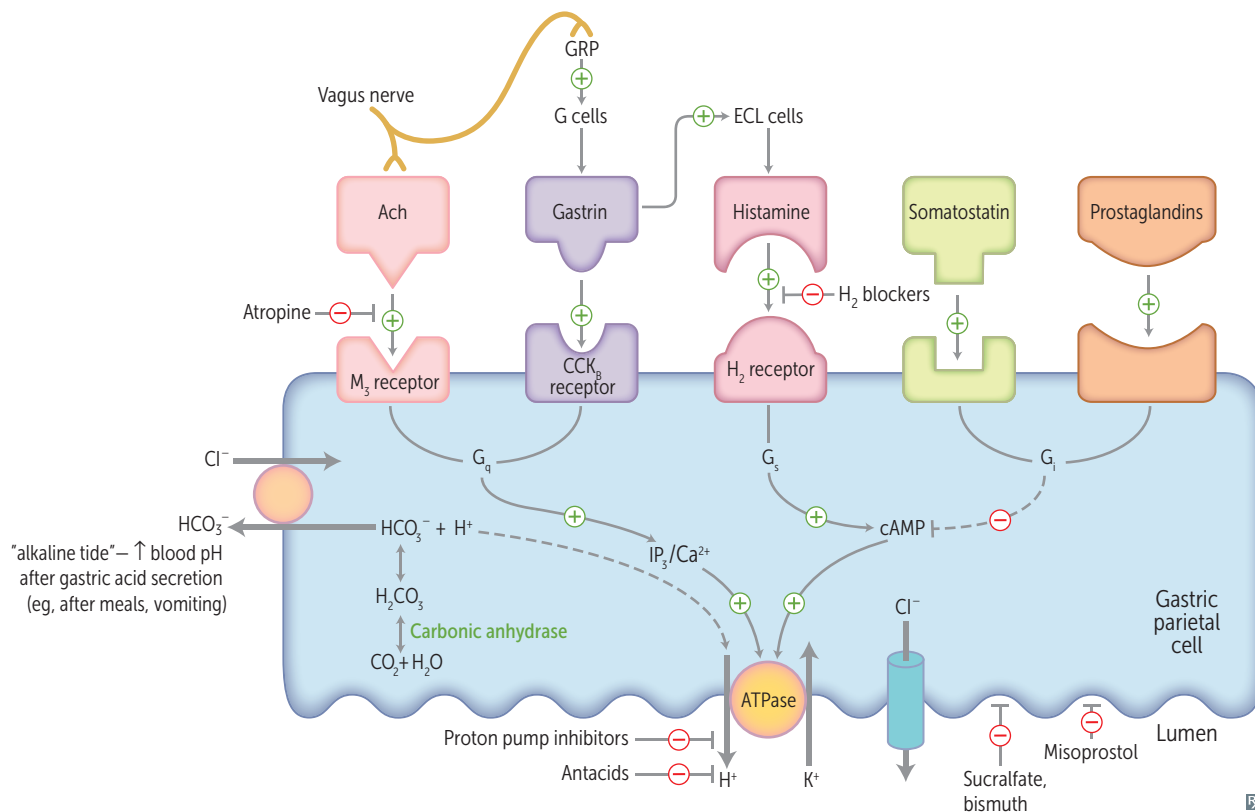
- Tobacco smoking (strongest risk factor)
- Chronic pancreatitis (especially > 20 years)
- Diabetes
- Age > 50 years

Often presents with:

- Abdominal pain radiating to back
- Weight loss (due to malabsorption and anorexia)
- Migratory thrombophlebitis—redness and tenderness on palpation of extremities (Trousseau syndrome)
- Obstructive jaundice with palpable, nontender gallbladder (Courvoisier sign)

► GASTROINTESTINAL—PHARMACOLOGY

Acid suppression therapy



H₂-blockers

Cimetidine, famotidine, nizatidine.

Take H₂ blockers before you dine. Think “table for 2” to remember H₂.

MECHANISM	Reversible block of histamine H ₂ -receptors → ↓ H ⁺ secretion by parietal cells.
CLINICAL USE	Peptic ulcer, gastritis, mild esophageal reflux.
ADVERSE EFFECTS	Cimetidine is a potent inhibitor of cytochrome P-450 (multiple drug interactions); it also has antiandrogenic effects (prolactin release, gynecomastia, impotence, ↓ libido in males); can cross blood-brain barrier (confusion, dizziness, headaches) and placenta. Cimetidine ↓ renal excretion of creatinine. Other H ₂ blockers are relatively free of these effects.

Proton pump inhibitors

Omeprazole, lansoprazole, esomeprazole, pantoprazole, dexlansoprazole.

MECHANISM	Irreversibly inhibit H ⁺ /K ⁺ -ATPase in stomach parietal cells.
CLINICAL USE	Peptic ulcer, gastritis, esophageal reflux, Zollinger-Ellison syndrome, component of therapy for <i>H pylori</i> , stress ulcer prophylaxis.
ADVERSE EFFECTS	↑ risk of <i>C difficile</i> infection, pneumonia, acute interstitial nephritis. Vitamin B ₁₂ malabsorption; ↓ serum Mg ²⁺ /Ca ²⁺ absorption (potentially leading to increased fracture risk in older adults).

Antacids

Can affect absorption, bioavailability, or urinary excretion of other drugs by altering gastric and urinary pH or by delaying gastric emptying. All can cause hypokalemia.

Aluminum hydroxide	Constipation, Hypophosphatemia, Osteodystrophy, Proximal muscle weakness, Seizures	Aluminum amount of feces CHOPS
Calcium carbonate	Hypercalcemia (milk-alkali syndrome), rebound acid ↑	Can chelate and ↓ effectiveness of other drugs (eg, tetracycline)
Magnesium hydroxide	Diarrhea, hyporeflexia, hypotension, cardiac arrest	Mg ²⁺ = Must go 2 the bathroom

Bismuth, sucralfate

MECHANISM	Bind to ulcer base, providing physical protection and allowing HCO ₃ ⁻ secretion to reestablish pH gradient in the mucous layer. Sucralfate requires acidic environment, not given with PPIs/H ₂ blockers.
CLINICAL USE	↑ ulcer healing, travelers' diarrhea (bismuth). Bismuth also used in quadruple therapy for <i>H pylori</i> .

Misoprostol

MECHANISM	PGE ₁ analog. ↑ production and secretion of gastric mucous barrier, ↓ acid production.
CLINICAL USE	Prevention of NSAID-induced peptic ulcers (NSAIDs block PGE ₁ production). Also used off-label for induction of labor (ripens cervix).
ADVERSE EFFECTS	Diarrhea. Contraindicated in patients of childbearing potential (abortifacient).

Octreotide

MECHANISM	Long-acting somatostatin analog; inhibits secretion of various splanchnic vasodilatory hormones.
CLINICAL USE	Acute variceal bleeds, acromegaly, VIPoma, carcinoid tumors.
ADVERSE EFFECTS	Nausea, cramps, steatorrhea. ↑ risk of cholelithiasis due to CCK inhibition.

Sulfasalazine

MECHANISM	A combination of sulfapyridine (antibacterial) and 5-aminosalicylic acid (anti-inflammatory). Activated by colonic bacteria.
CLINICAL USE	Ulcerative colitis, Crohn disease (colitis component).
ADVERSE EFFECTS	Malaise, nausea, sulfonamide toxicity, reversible oligospermia.

Loperamide, diphenoxylate

MECHANISM	Agonists at μ -opioid receptors → ↓ gut motility. Poor CNS penetration (low addictive potential).
CLINICAL USE	Diarrhea.
ADVERSE EFFECTS	Constipation, nausea.

Antiemetics

All act centrally in chemoreceptor trigger zone of area postrema.

DRUG	MECHANISM	CLINICAL USE	ADVERSE EFFECTS
Ondansetron, granisetron	5-HT ₃ -receptor antagonists Also act peripherally (↓ vagal stimulation)	Nausea and vomiting after chemotherapy, radiotherapy, or surgery	Headache, constipation, QT interval prolongation, serotonin syndrome
Prochlorperazine, metoclopramide	D ₂ -receptor antagonists Metoclopramide also causes ↑ gastric emptying and ↑ LES tone	Nausea and vomiting Metoclopramide is also used in gastroparesis (eg, diabetic), persistent GERD	Extrapyramidal symptoms, hyperprolactinemia, anxiety, drowsiness, restlessness, depression, GI distress
Aprepitant, fosaprepitant	NK ₁ (neurokinin-1) receptor antagonists NK ₁ receptor = substance P receptor	Chemotherapy-induced nausea and vomiting	Fatigue, GI distress

Orlistat

MECHANISM	Inhibits gastric and pancreatic lipase → ↓ breakdown and absorption of dietary fats. Taken with fat-containing meals.
CLINICAL USE	Weight loss.
ADVERSE EFFECTS	Abdominal pain, flatulence, bowel urgency/frequent bowel movements, steatorrhea; ↓ absorption of fat-soluble vitamins.

Anticonstipation drugs

DRUG	MECHANISM	ADVERSE EFFECTS
Bulk-forming laxatives Methylcellulose, psyllium	Soluble fibers that draw water into gut lumen, forming viscous liquid that promotes peristalsis	Bloating
Osmotic laxatives Lactulose, magnesium citrate, magnesium hydroxide, polyethylene glycol	Provide osmotic load to draw water into GI lumen Lactulose also treats hepatic encephalopathy: gut microbiota degrades lactulose into metabolites (lactic acid, acetic acid) that promote nitrogen excretion as NH_4^+ by trapping it in colon	Diarrhea, dehydration; may be misused by patients with bulimia
Stimulant laxatives Bisacodyl, senna	Enteric nerve stimulation → colonic contraction	Diarrhea
Emollient laxatives Docusate	Surfactants that ↓ stool surface tension, promoting water entry into stool	Diarrhea
Lubiprostone	Chloride channel activator → ↑ intestinal fluid secretion	Diarrhea, nausea
Guanylate cyclase-C agonists Linaclotide, plecanatide	Activate intracellular cGMP signaling → ↑ fluid and electrolyte secretion in the intestinal lumen	Diarrhea, bloating, abdominal discomfort, flatulence
Serotonergic agonists Prucalopride	5HT_4 agonism → enteric nerve stimulation → ↑ peristalsis, intestinal secretion	Diarrhea, abdominal pain, nausea, headache
NHE_3 inhibitor Tenapanor	Inhibits Na^+/H^+ exchanger → ↓ Na^+ absorption → ↑ H_2O secretion in lumen	Diarrhea, abdominal pain, nausea